

Changyu Zhao

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EDUCATION

University of Wisconsin-Madison

B.S. in Computer Science (Honors) and Mathematics

Cumulative GPA: **3.97/4.0**, Dean's List 2021 - 2024

Sep. 2021 – May 2025

Madison, Wisconsin

EXPERIENCES

UW-Madison Cybersecurity Operation Center

Cybersecurity Student Intern Analyst

Feb. 2023 – Present

Madison, Wisconsin

- Managed 400+ daily campus security incidents: respond to alerts, investigate security breaches, and take corrective action.
- Utilized **Elastic** and **Palo Alto Firewall** to **monitor and analyze campus network traffic**, identified potential threats and vulnerabilities, and notify security alerts to relevant campus users or departments.
- Conducted **Identity and Access Management (IAM)** compliance audit process for 200+ campus employee accounts.
- Collaborated with team members to **process daily threat intel**, enhancing the campus security posture.

ASU SEFCOM/CDF Security Research Apprenticeship

Research Assistant

May. 2023 – Jul. 2023 & May. 2024 – Sep. 2024

Tempe, Arizona

- Developed **LibHopper**, a Python-based tool to **automate discovery of exploitation primitives** in user libraries.
- Leveraged automated processes to **pipeline the evaluation of security implications** related to memory corruption vulnerabilities in library internal structures, enhancing the capabilities of security researchers and developers.
- Customized debugger scripts and used **symbolic execution techniques** for comprehensive **exploitability analysis**.
- Cataloged **exploitable primitives** from analysis of the 30 most widely-used C standard libraries.

Tencent Security Xuanwu Lab

Security Research Intern

Jul. 2023 – Aug. 2023

Beijing, China

- Developed **LibIHT**, a cross-platform, hardware-assisted dynamic control flow analysis tool in **C** and **C++** that leverages Intel's **Last Branch Record (LBR)** and **Branch Trace Store (BTS)** hardware tracing features.
- Designed and implemented **kernel modules** and **device drivers** to enable tracing features for Linux and Windows.
- Architected a user-mode analysis library to **accurately reconstruct control flow graphs (CFG)** from trace data, enhancing program tracing and profiling capabilities.

PROJECTS

Router automated vulnerability discovery | *Reverse Engineering, IoT, Python*

May 2022 – Aug. 2022

- Discovered and reported 7 **unfixed Common Vulnerabilities and Exposures (CVE)** in Tenda AC 18 router.
- Developed an **automated IDA Pro plugin** for detecting **stack overflow and command injection vulnerabilities**.
- Utilized **QEMU** to **rehost the ARM-based router firmware** on x86-64 machine for in-depth debugging and analysis.
- Demonstrated the ability to **achieve Remote Code Execution (RCE)** by exploiting the identified vulnerabilities.

EXTRACURRICULAR ACTIVITIES

r3kapig CTF Team

Team member

Sep. 2021 – Present

China

- Achieved a notable accomplishment by securing a rank 12/1282 worldwide in the DEFCON 30 CTF Finals (2022), and 7/1828 worldwide in the DEFCON 31 CTF Finals (2023).
- Participated in over 40 48-hour Capture the Flag (CTF) competitions since 2021, with a focus on Binary Exploitation (PWN) and Reverse Engineering challenges.

pwn.college

Independent Learning

Dec. 2021 – Jul. 2022

Tempe, Arizona

- Attained the Yellow Belt and Blue Belt levels by completing **all** challenges in Dojo instance, currently ranked 43/2500+.
- Acquired comprehensive skills in various areas of **software and system security**, including program interaction, shellcoding, privilege escalation, reverse engineering, stack corruption, heap corruption, race conditions, and kernel security.

TECHNICAL SKILLS

Proficient in reverse engineering, have reverse experience in large-scale projects. **Familiar** with control flow tracking, data flow tracking, symbolic execution and the use of binary analysis tools.

Programming Languages: C/C++, Python, Java, Assembly Language, JavaScript, HTML/CSS, LaTeX

Binary Security Tools: IDA Pro/Ghidra, GDB/Windbg, Angr, Qemu, Frida, Joern

Web Security Tools: Burpsuit, Wireshark, Metasploit Framework, Kali Linux

Soft Skills: Time/Project Management, Collaboration, Conflict Resolution, Active Listening